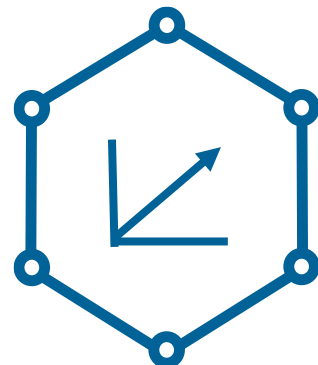


Knowledge Graphs

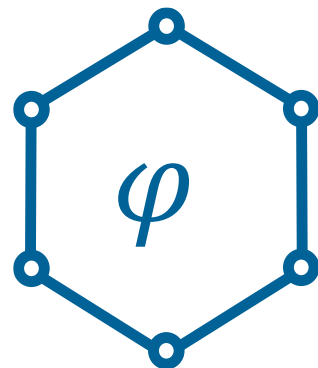


Emanuel Sallinger



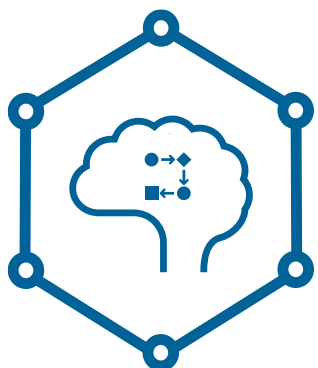
KG Embeddings

Widely-applied, large family of ML models



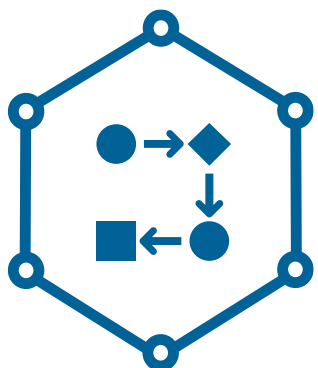
Logical Knowledge in KGs

Highly expressive, diverse family of logical models.



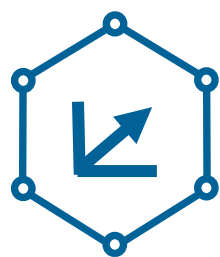
Graph Neural Networks

Using the KG structure *as* a neural network.

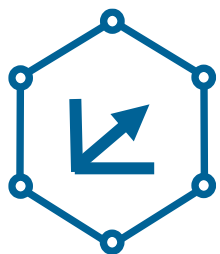


Data Models

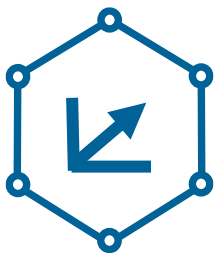
Overview of data models in different communities



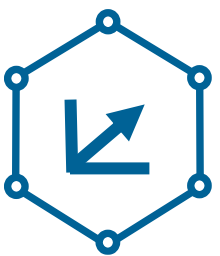
Knowledge Graph Embeddings



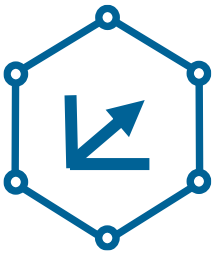
Knowledge Graph Embeddings
TransE



TransE
Learning

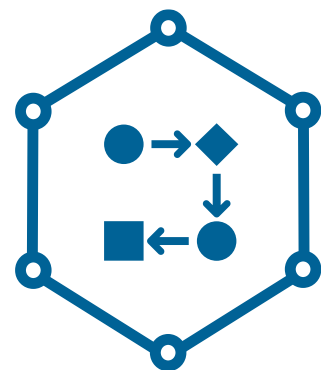


Knowledge Graph Embeddings
A Glimpse Beyond



Knowledge Graph Embeddings
Theory to Practice





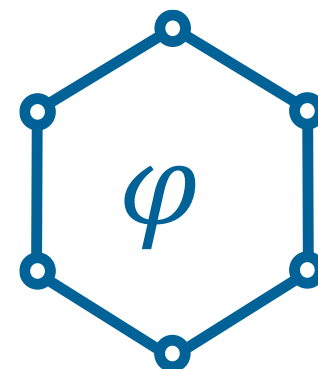
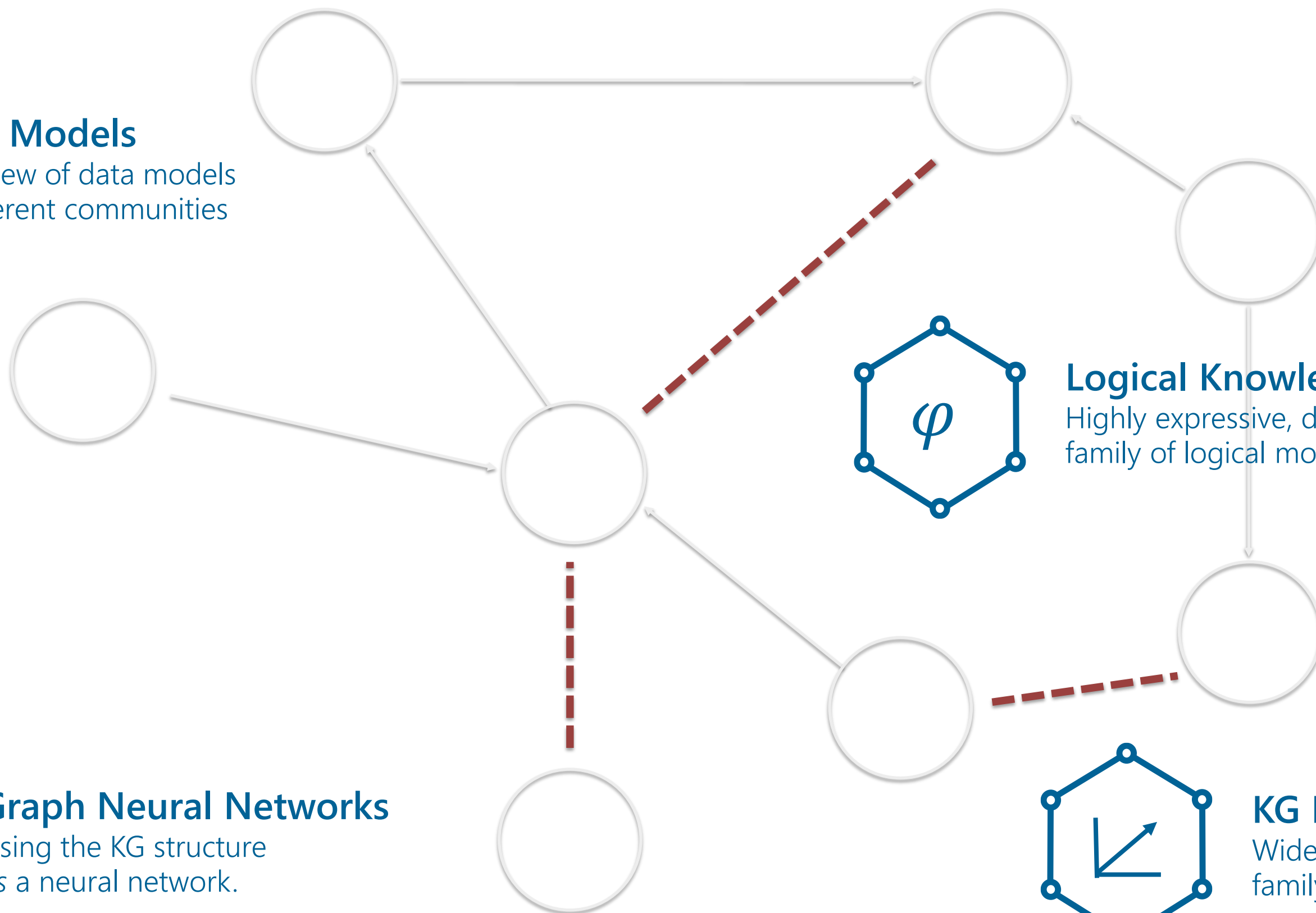
Data Models

Overview of data models in different communities



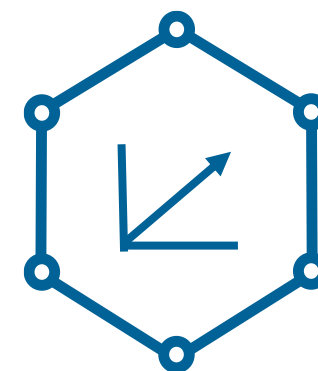
Graph Neural Networks

Using the KG structure as a neural network.



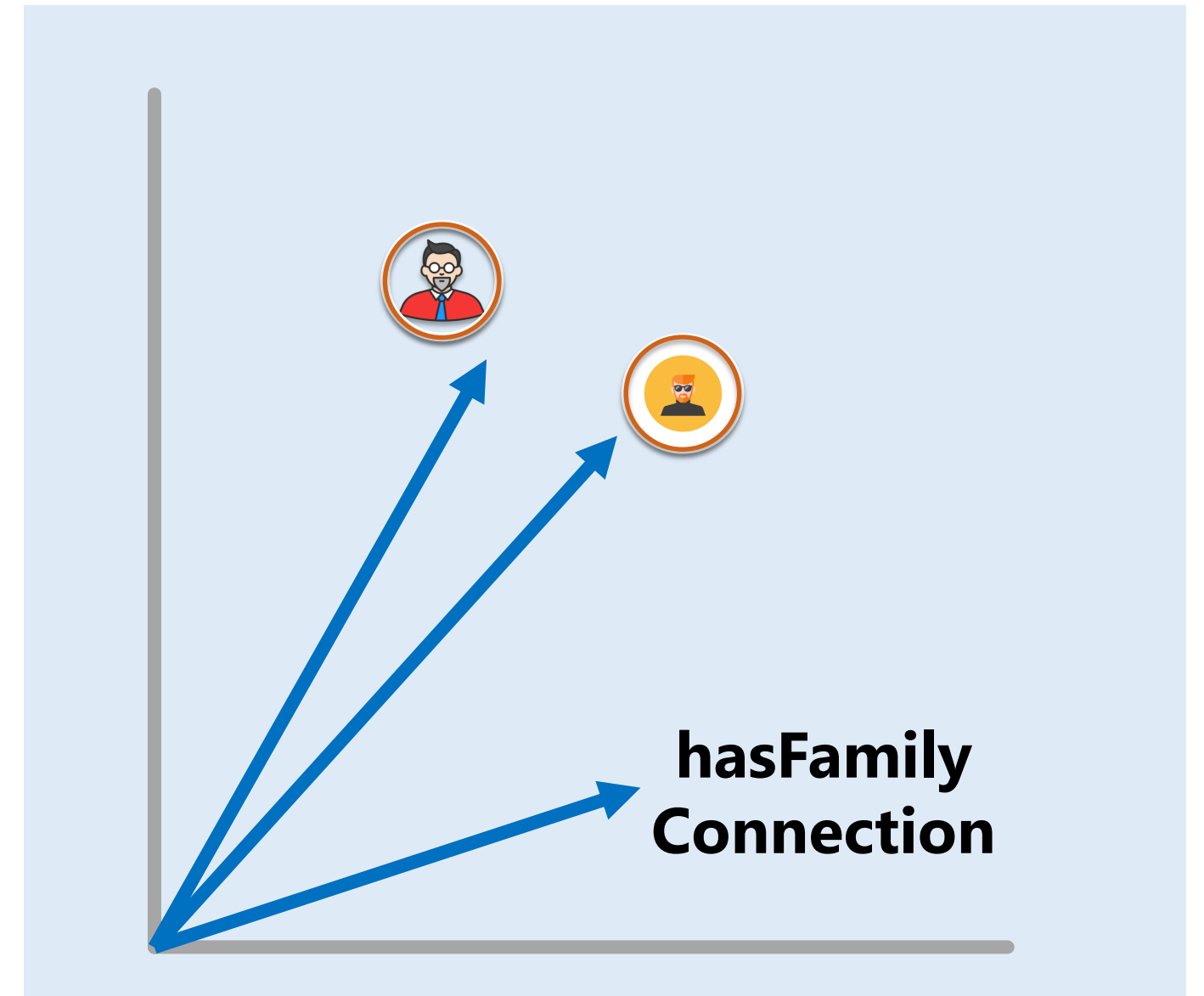
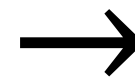
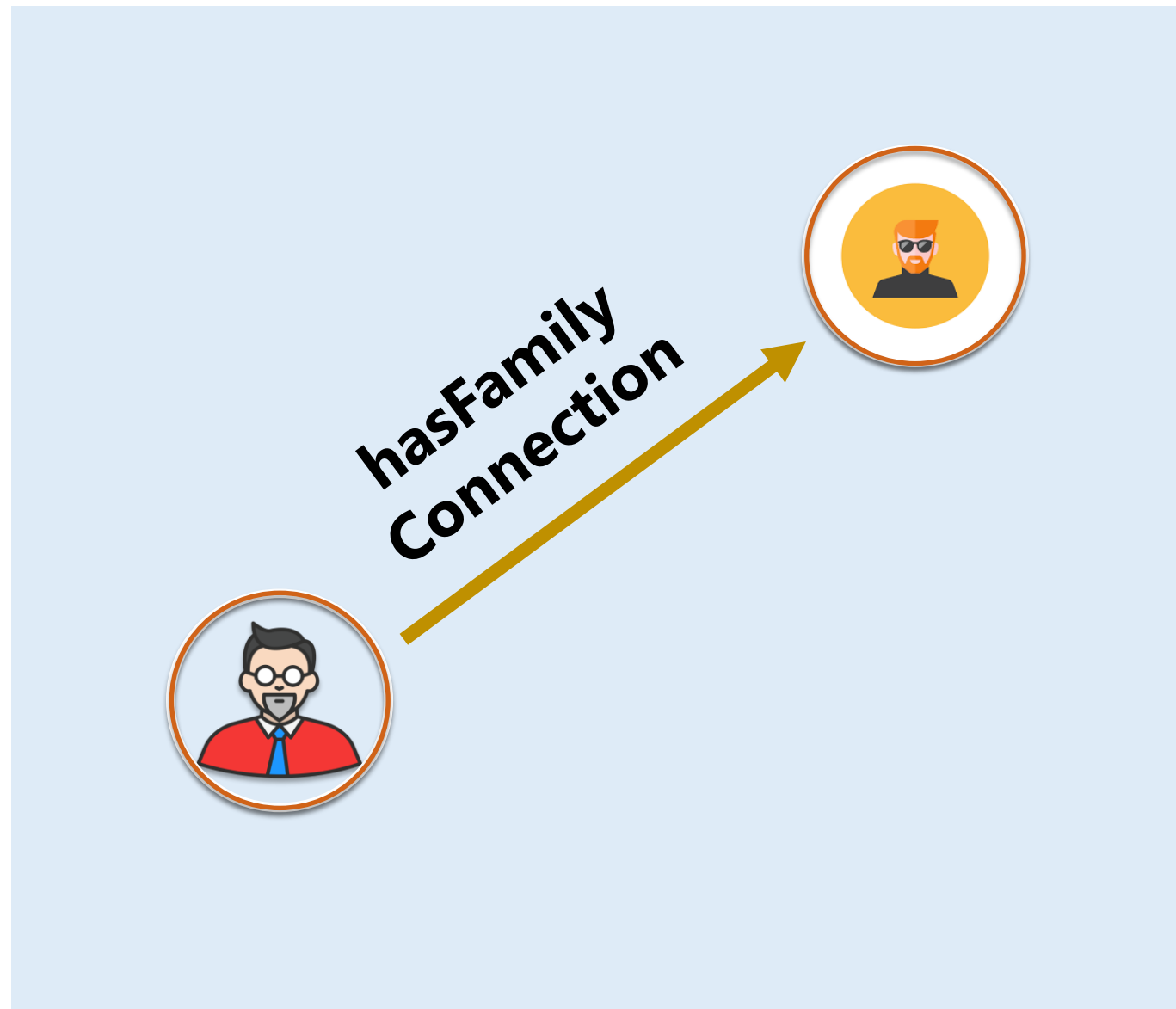
Logical Knowledge in KGs

Highly expressive, diverse family of logical models.



KG Embeddings

Widely-applied, large family of ML models



Knowledge Graph Embeddings:

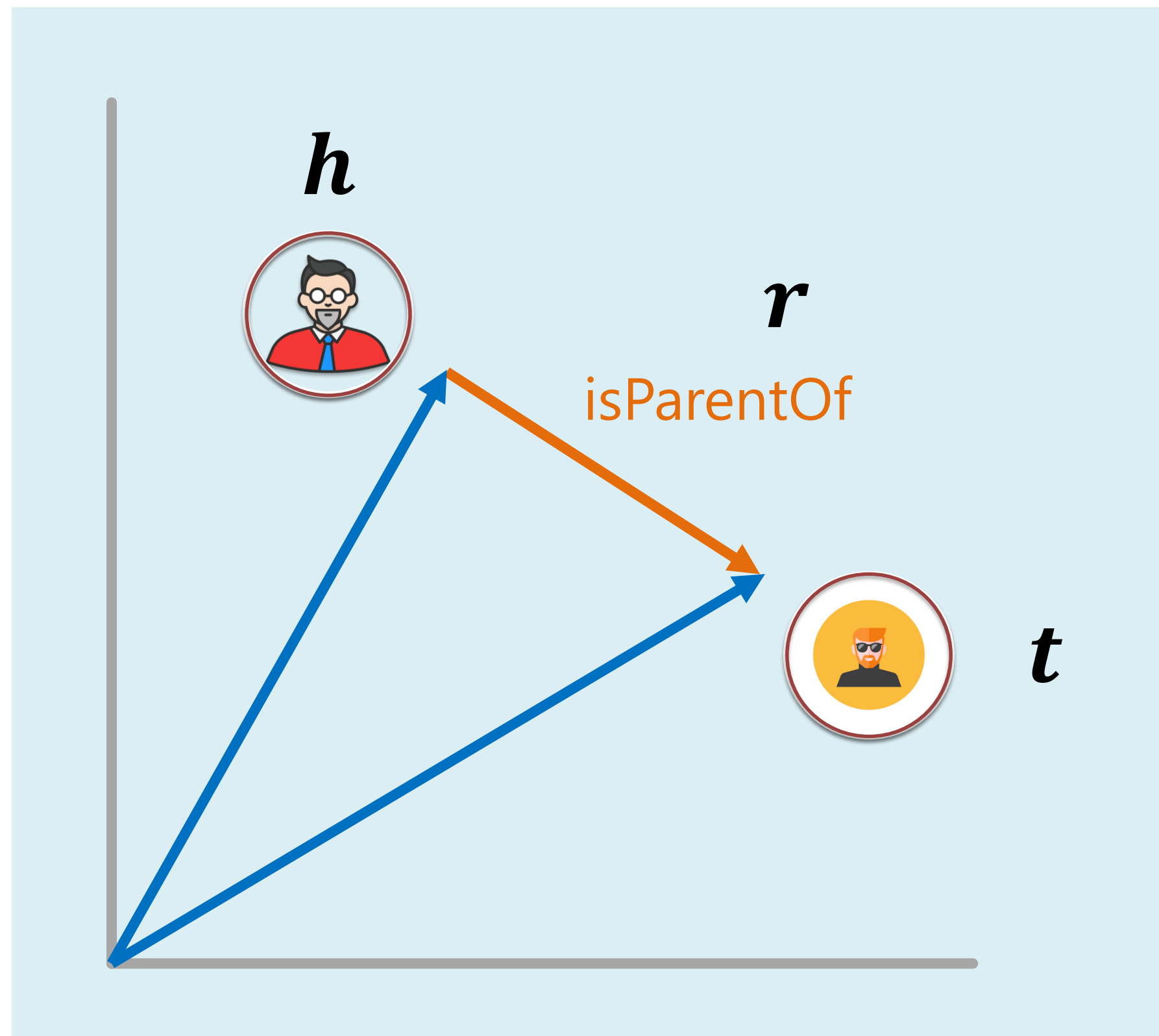
Latent Knowledge

$$KG = (D, K; g, l, r)$$

$$\mathbb{K} = \mathbb{e}$$

$$l: \mathbb{D} \rightarrow \mathbb{e}$$

$$r_{\mathbb{e}}: \mathbb{D} \rightarrow \mathbb{D}$$



$$h + r \approx t$$



KG Embeddings

Source: <https://github.com/pykeen/pykeen>

Datasets (25)				
The citation for each dataset corresponds to either the paper describing the dataset, the first paper published using the dataset with knowledge graph embedding models, or the URL for the dataset if neither of the first two are available.				
Name	Documentation	Citation	Entities	Relations
Clinical Knowledge Graph	<code>pykeen.datasets.CKG</code>	Santos et al., 2020	7617419	11
CoDEx (large)	<code>pykeen.datasets.CoDExLarge</code>	Safavi et al., 2020	77951	69
CoDEx (medium)	<code>pykeen.datasets.CoDExMedium</code>	Safavi et al., 2020	17050	51
CoDEx (small)	<code>pykeen.datasets.CoDExSmall</code>	Safavi et al., 2020	2034	42
ConceptNet	<code>pykeen.datasets.ConceptNet</code>	Speer et al., 2017	28370083	50
Countries	<code>pykeen.datasets.Countries</code>	ZhenfengLei/KGDatasets	271	2
Commonsense Knowledge Graph	<code>pykeen.datasets.CSKG</code>	Ilievski et al., 2020	2087833	58
DB100K	<code>pykeen.datasets.DB100K</code>	Ding et al., 2018	99604	470
DBpedia50	<code>pykeen.datasets.DBpedia50</code>	Shi et al., 2017	24624	351
Drug Repositioning Knowledge Graph	<code>pykeen.datasets.DRKG</code>	gnn4dr/DRKG	97238	107
FB15k	<code>pykeen.datasets.FB15k</code>	Bordes et al., 2013	14951	1345
FB15k-237	<code>pykeen.datasets.FB15k237</code>	Toutanova et al., 2015	14505	237
Hetionet	<code>pykeen.datasets.Hetionet</code>	Himmelstein et al., 2017	45158	24
Kinships	<code>pykeen.datasets.Kinships</code>	ZhenfengLei/KGDatasets	104	25
Nations	<code>pykeen.datasets.Nations</code>	ZhenfengLei/KGDatasets	14	55
OGB BioKG	<code>pykeen.datasets.OGBBioKG</code>	Hu et al., 2020	45085	51
OGB WikiKG	<code>pykeen.datasets.OGBWikiKG</code>	Hu et al., 2020	2500604	535
OpenBioLink	<code>pykeen.datasets.OpenBioLink</code>	Breit et al., 2020	180992	28
OpenBioLink (F1)	<code>pykeen.datasets.OpenBioLinkF1</code>	PyKEEN/pykeen-openbiolink-benchmark	116425	19
OpenBioLink (F2)	<code>pykeen.datasets.OpenBioLinkF2</code>	PyKEEN/pykeen-openbiolink-benchmark	110628	17

Models (25)		
Name	Reference	Citation
ComplEx	<code>pykeen.models.ComplEx</code>	Trouillon et al., 2016
ComplExLiteral	<code>pykeen.models.ComplExLiteral</code>	Kristiadi et al., 2018
ConvE	<code>pykeen.models.ConvE</code>	Dettmers et al., 2018
ConvKB	<code>pykeen.models.ConvKB</code>	Nguyen et al., 2018
DistMult	<code>pykeen.models.DistMult</code>	Yang et al., 2014
DistMultLiteral	<code>pykeen.models.DistMultLiteral</code>	Kristiadi et al., 2018
ERMLP	<code>pykeen.models.ERMLP</code>	Dong et al., 2014
ERMLPE	<code>pykeen.models.ERMLPE</code>	Sharifzadeh et al., 2019
HoIE	<code>pykeen.models.HoIE</code>	Nickel et al., 2016
KG2E	<code>pykeen.models.KG2E</code>	He et al., 2015
MuRE	<code>pykeen.models.MuRE</code>	Balažević et al., 2019
NTN	<code>pykeen.models.NTN</code>	Socher et al., 2013
PairRE	<code>pykeen.models.PairRE</code>	Chao et al., 2020
ProjE	<code>pykeen.models.ProjE</code>	Shi et al., 2017
RESCAL	<code>pykeen.models.RESCAL</code>	Nickel et al., 2011
RGCN	<code>pykeen.models.RGCN</code>	Schlichtkrull et al., 2018
RotatE	<code>pykeen.models.RotatE</code>	Sun et al., 2019
Simple	<code>pykeen.models.Simple</code>	Kazemi et al., 2018
StructuredEmbedding	<code>pykeen.models.StructuredEmbedding</code>	Bordes et al., 2011
TransD	<code>pykeen.models.TransD</code>	Ji et al., 2015
TransE	<code>pykeen.models.TransE</code>	Bordes et al., 2013



Algorithm 1 Learning TransE

input Training set $S = \{(h, \ell, t)\}$, entities and rel. sets E and L , margin γ , embeddings dim. k .

```
1: initialize  $\ell \leftarrow \text{uniform}$ 
2:    $\ell \leftarrow \ell / \|\ell\|$ 
3:    $e \leftarrow \text{uniform}$ 
4: loop
5:    $e \leftarrow e / \|e\|$ 
6:    $S_{batch} \leftarrow \text{sample}$ 
7:    $T_{batch} \leftarrow \emptyset$  // in
8:   for  $(h, \ell, t) \in S_{batch}$  do
9:      $(h', \ell, t') \leftarrow \text{sample}(S'_{(h, \ell, t)})$  // sample a corrupted triplet
10:     $T_{batch} \leftarrow T_{batch} \cup \{((h, \ell, t), (h', \ell, t'))\}$ 
11:   end for
12:   Update embeddings w.r.t.  $\sum_{((h, \ell, t), (h', \ell, t')) \in T_{batch}} \nabla [\gamma + d(\mathbf{h} + \ell, \mathbf{t}) - d(\mathbf{h}' + \ell, \mathbf{t}')]_+$ 
13: end loop
```

Negative Samplers (2)

Name	Reference	Description
basic	<code>pykeen.sampling.BasicNegativeSampler</code>	A basic negative sampler.
bernoulli	<code>pykeen.sampling.BernoulliNegativeSampler</code>	An implementation of the Bernoulli negative sampling approach proposed by [wang2014].

Negative sample

Stochastic
Gradient Descent
update phase